UNIVERSITY INSTITUTE OF COMPUTING

DIVISION- MCA/BCA/BSc(CS)

Chandigarh University Campus Network

A Project Report

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Submitted by:

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IN

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UNIVERSITY INSTITUTE OF COMPUTING

DIVISION- MCA/BCA/BSc(CS)

BONAFIDE CERTIFICATE

Certified that this project report "**Chandigarh University Campus Network"** is the Bonafide work of "**Gaurav Raj"** who carried out the project work under my/our supervision.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

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ABSTRACT

This project presents the design and implementation of a secure campus network for Chandigarh University using Cisco Packet Tracer. The objective of the project is to create a departmental network architecture where multiple departments such as Admin Block, Faculty Block, Student Lab, and Server Room communicate efficiently through VLAN-based segmentation.

The network is designed using switches, routers, servers, and end devices. VLAN technology is implemented to logically separate departments and reduce broadcast traffic. Inter-VLAN routing is achieved through router-on-stick configuration so that departments can communicate according to defined policies.

Dynamic Host Configuration Protocol (DHCP) is configured using the router to automatically assign IP addresses to all PCs. Domain Name System (DNS) service is implemented using the domain cu.edu, enabling hostname-based communication. Internal email service is also configured for communication between departments. Access Control Lists (ACLs) are applied to restrict student users from accessing the Admin network.

The project demonstrates practical understanding of enterprise network design, security policies, and server integration in a campus environment.

CHAPTER 1
INTRODUCTION

1.1 Need of the Project

Educational institutions require reliable communication networks for sharing resources, accessing servers, and ensuring secure communication between departments.
A campus network helps in:

- connecting multiple departments
- reducing manual configuration
- improving internal communication
- increasing security

Traditional flat LAN networks create large broadcast domains and weak security.

1.2 Problem Identification

Without segmentation:

- students may access admin systems
- network congestion increases
- IP conflicts occur
- server services become difficult to manage

Therefore VLAN-based network design becomes necessary.

1.3 Objectives of the Project

Main objectives:

- create separate VLANs for departments
- configure inter-VLAN routing
- provide automatic IP assignment
- configure DNS, Web and Email servers
- apply ACL for access restriction

1.4 Scope of Project

The project covers:

- Admin Department
- Faculty Department
- Student Lab
- Server Room

Each section operates in separate logical network.

CHAPTER 2: BACKGROUND STUDY

2.1 VLAN Concept
A VLAN (Virtual Local Area Network) divides a physical network into multiple logical networks.
Advantages:
- reduces broadcast traffic
- improves security
- simplifies management

2.2 Inter-VLAN Routing
Different VLANs cannot communicate directly.
A router is used for communication through subinterfaces.
Router-on-stick method is used.

2.3 DHCP
DHCP automatically assigns:

- IP address
- subnet mask
- gateway
- DNS

This removes manual configuration.

2.4 DNS
DNS converts domain names into IP addresses.
Domain used:
cu.edu

2.5 ACL Security
ACL controls communication between departments.
Student VLAN is restricted from Admin VLAN.

CHAPTER 3: SYSTEM DESIGN

3.1 Network Devices Used

Devices used:

- 1 Router
- 5 Switches
- 1 DNS Server
- 1 Email Server
- 1 Web Server
- PCs

3.2 Department VLAN Design

VLAN	Department	Network
10	Admin	192.168.10.0/24
20	Faculty	192.168.20.0/24
30	Student	192.168.30.0/24
40	Server	192.168.40.0/24

3.3 Topology Design

The topology consists of:

- Core switch at center
- Department switches connected to core switch
- Router connected for routing
- Servers connected to server switch

(Insert topology screenshot here)

3.4 IP Addressing Plan

Device	IP
Router VLAN10	192.168.10.1
Router VLAN20	192.168.20.1
Router VLAN30	192.168.30.1
Router VLAN40	192.168.40.1

CHAPTER 4: IMPLEMENTATION

4.1 VLAN Configuration

```
Switch>
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name ADMIN
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name FACULTY
Switch(config-vlan)#vlan 30
Switch(config-vlan)#name STUDENTS
Switch(config-vlan)#vlan 40
Switch(config-vlan)#name SERVERS
Switch(config-vlan)#
```

4.2 Trunk Configuration

Trunk links connect switches:

```
switchport mode trunk
```

4.3 Router Configuration

Sub interfaces created:

```
interface g0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
```

4.4 DHCP Configuration

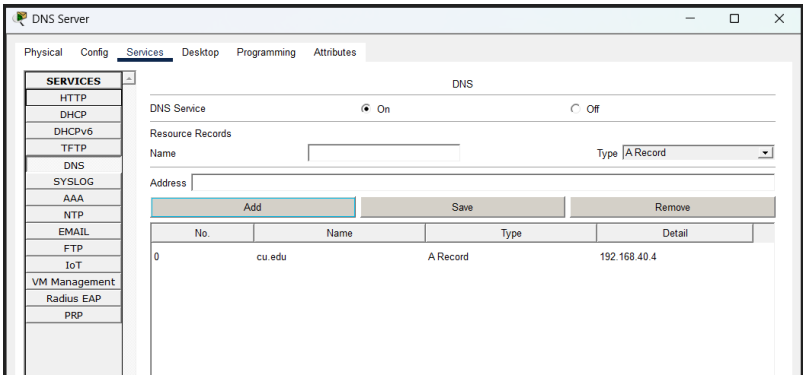
Router DHCP pools:

```
ip dhcp pool ADMIN
network 192.168.10.0 255.255.255.0
default-router 192.168.10.1
dns-server 192.168.40.2
```

4.5 DNS Configuration

DNS records:

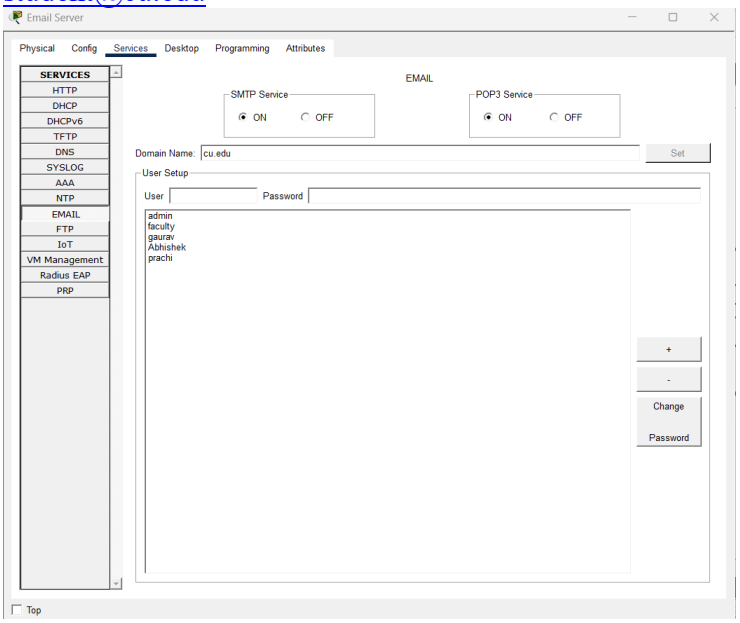
- cu.edu
- mail.cu.edu



4.6 Email Server Configuration

Email accounts created:

- admin@cu.edu
- faculty@cu.edu
- student@cu.edu

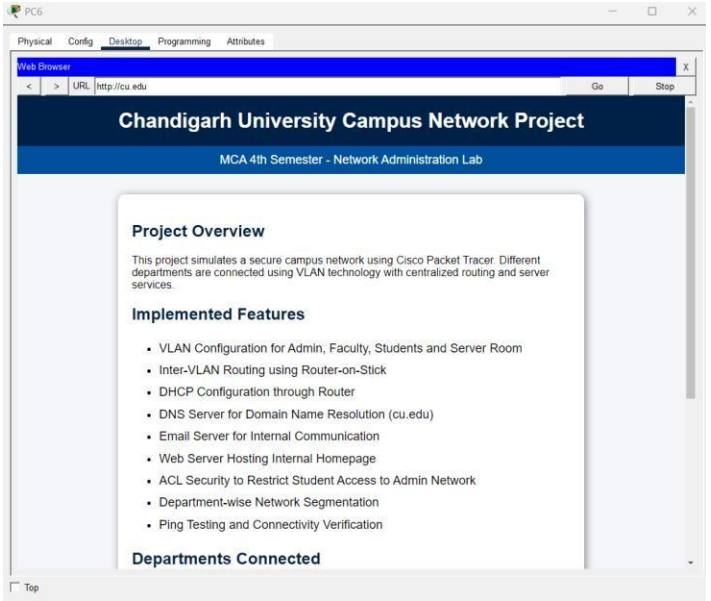


4.7 Web Server Configuration

Internal homepage hosted.

Accessible by browser:

cu.edu



4.8 ACL Configuration

Student restriction:

```
access-list 100 deny ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255
access-list 100 permit ip any any
```

CHAPTER 5: TESTING AND RESULT

5.1 VLAN Connectivity Test

Ping successful between:

- Admin ↔ Faculty

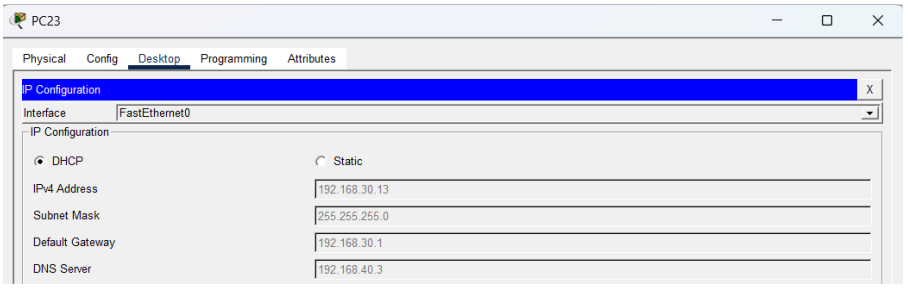
```
C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Reply from 192.168.20.2: bytes=32 time=2ms TTL=127
Reply from 192.168.20.2: bytes=32 time=1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time=16ms TTL=127

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 16ms, Average = 4ms

C:\>
```

- Faculty ↔ Student

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.30.13

Pinging 192.168.30.13 with 32 bytes of data:

Reply from 192.168.30.13: bytes=32 time<1ms TTL=127
Reply from 192.168.30.13: bytes=32 time=10ms TTL=127
Reply from 192.168.30.13: bytes=32 time<1ms TTL=127
Reply from 192.168.30.13: bytes=32 time=3ms TTL=127

Ping statistics for 192.168.30.13:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 3ms

C:\>
```

5.2 DHCP Test

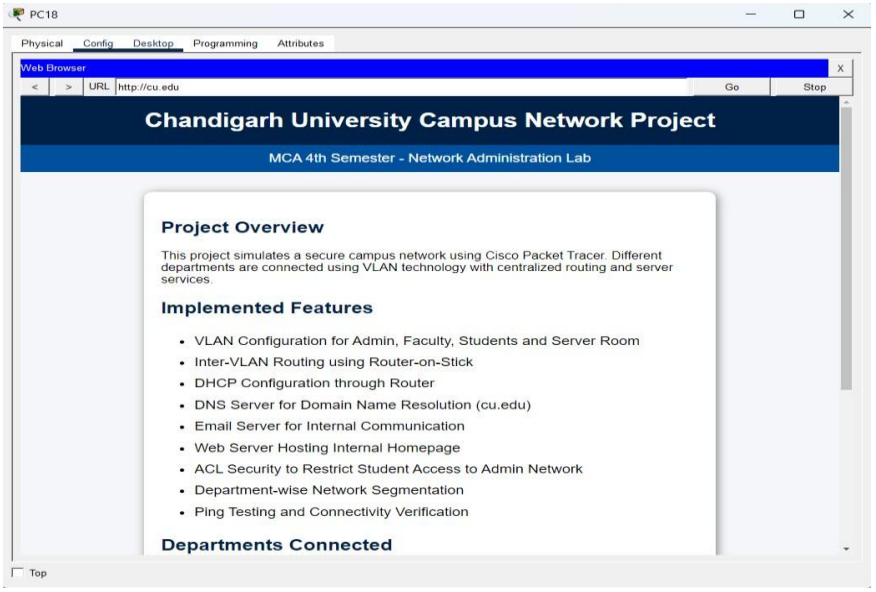
All PCs received automatic IP.



5.3 DNS Test

Browser resolved:

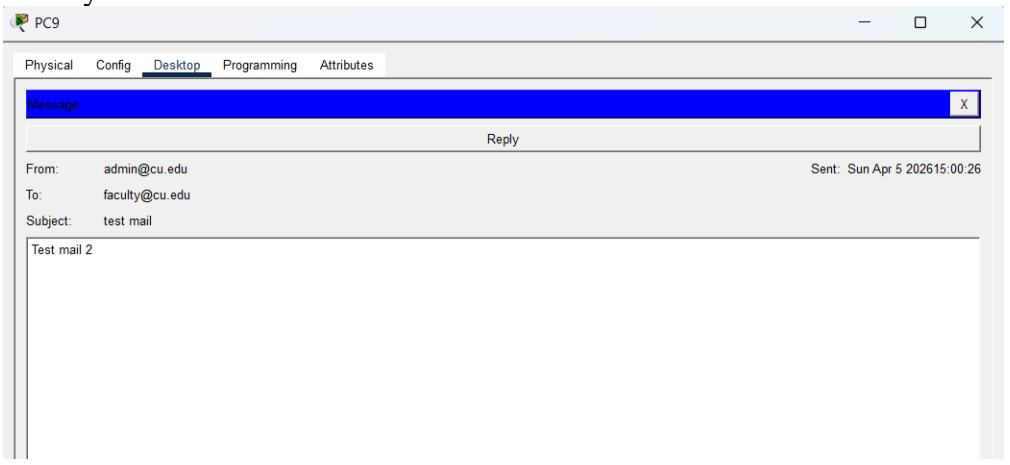
cu.edu



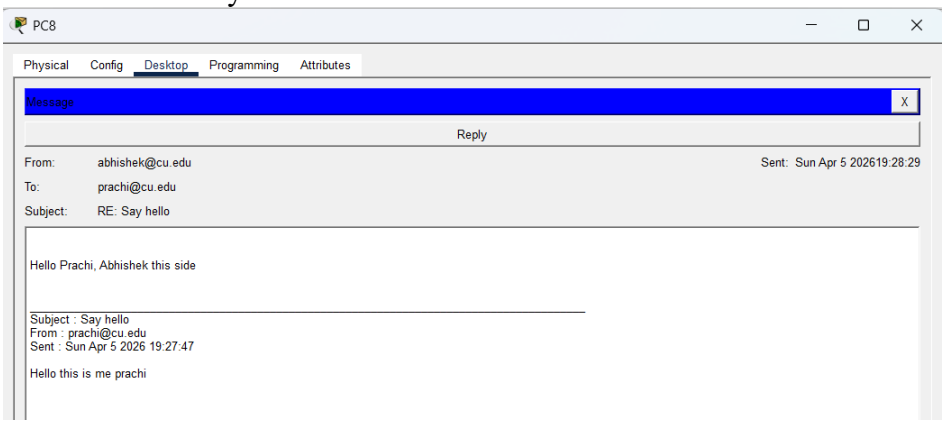
5.4 Email Test

Mail successfully sent:

Faculty → Student



Student → Faculty

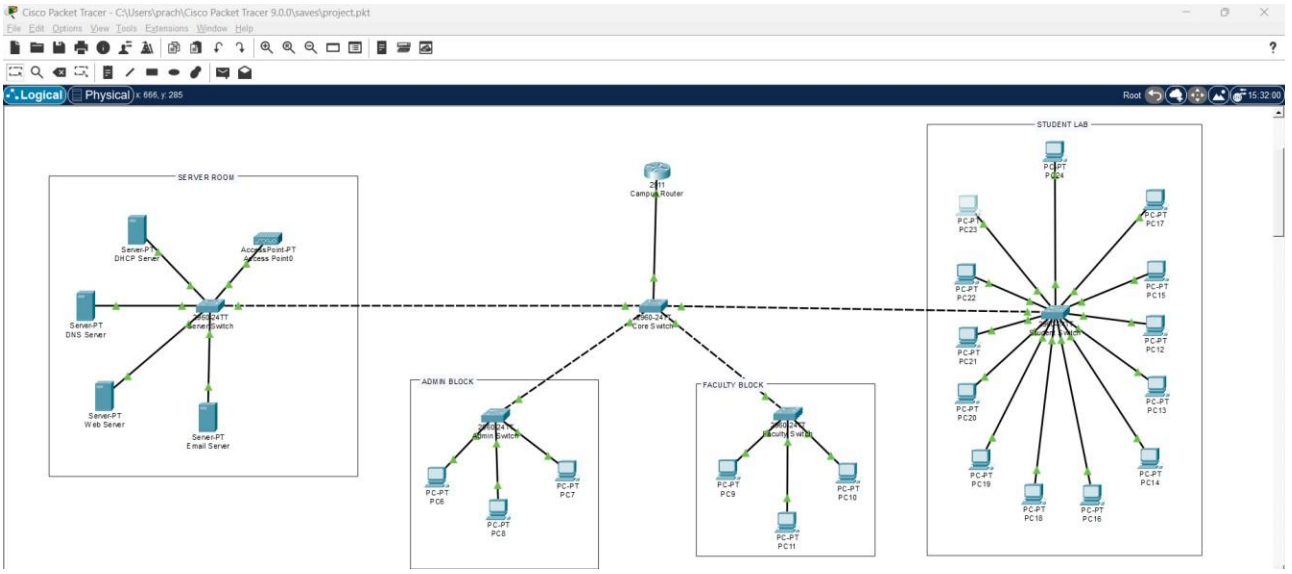


5.5 ACL Result

Student → Admin blocked successfully.

This proves access control works.

```
Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```



CONCLUSION

The Chandigarh University Campus Network Project successfully demonstrates practical campus network design using Cisco Packet Tracer.

Implemented technologies include:

- VLAN
- Routing
- DHCP
- DNS
- Email
- Web
- ACL Security

The project simulates real enterprise network administration.

FUTURE ENHANCEMENT

Future upgrades:

- wireless network
- firewall integration
- separate DHCP server
- network monitoring